Understanding Unix

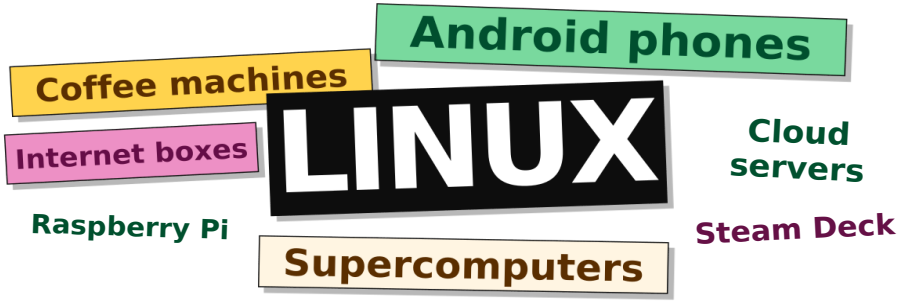
NUnix · L3 MIAGE · CM1 · 90 minutes

ISTIC · University of Rennes 2026–2027

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Linux is closer than you think

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You can use it all day without opening a terminal.

Yes, even the coffee machine

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Android phones

The **Linux kernel** manages processes, memory and hardware.

Google · platform architecture



Freebox Server Pop

A home internet box. Free publishes its **Linux source code**.

Free · firmware 4.9.10



MARO Model One

A coffee machine with **embedded Linux**, remote support and updates.

Toradex · May 2026

Linux can run behind a touchscreen or a web interface.

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Unix and Linux usage

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Mobile web use

69.17%

of observed mobile page views come from **Android**, built on Linux.

Statcounter · Sept 2026

Websites

92.1%

use the **Unix family**, including Linux, among sites whose OS W3Techs identifies.

W3Techs · 1 Oct 2026

Supercomputers

500/500

systems in the **June 2026 TOP500 list** use the Linux family.

TOP500 · June 2026

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$ whoami

Who teaches NUnix, and how is the course organised?

5/56

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Associate Professor
University of Rennes · ISTIC
IRISA / Inria · Taran

- Arithmetic circuits: **MLIR (LLVM) / CIRCT**.
- **Systolic arrays** and small floating-point formats.

Previously: Inria / INSA Lyon, Emeraude.
PhD: **BSC / UPC**.
bynaryman.github.io



MareNostrum 4 · BSC, Barcelona

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CM, TD and TP

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NUnix · L3 MIAGE · 2026–2027

Sessions	Teaching team
CM · lectures	Louis Ledoux
TD · tutorials	Elisabeth Bouriah and Louis Ledoux
TP · practical sessions	Elisabeth Bouriah and Louis Ledoux

In TD and TP, we practise the lecture topics through exercises, commands and Bash scripts.

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Course structure

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Three lectures (CM) of 90 minutes.

Lecture	Topics	Application
CM1 · Unix	Files, paths and permissions	Find files and check access
CM2 · Commands	Processes, search, pipes and redirections	Count errors in a log
CM3 · Bash	Variables, conditions, loops and functions	Write scripts with arguments and loops

TD and TP: practise commands first, then write scripts.

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One line, a useful result

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Which errors occur most often in **1,000 log entries**?

```
sort < errors.log | uniq -c | sort -nr | tee output.txt |
  a2ps -1 -o - | ps2pdf - rapport.pdf
```

Save the counts in `output.txt` and the report in `rapport.pdf`.

errors.log · first three lines

```
Database timeout
Login rejected
Connection refused
```

output.txt · first three counts

```
    320 Database timeout
    210 Login rejected
    140 Connection refused
```

a2ps: text → PostScript
ps2pdf: PostScript → PDF

By the end of this course, you will understand this command.

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1. Linux and the shell

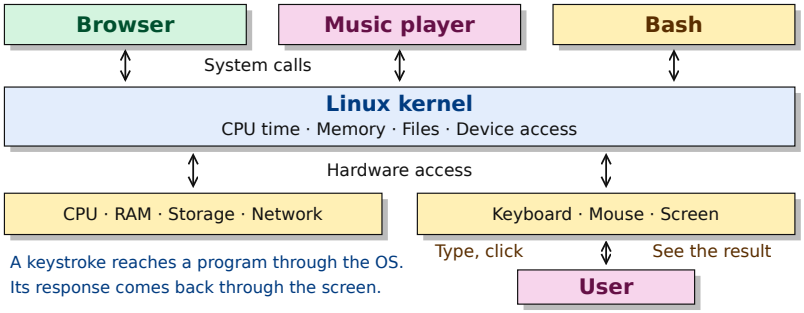
What happens between typing a command and the machine doing the work?

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Programs, kernel and hardware

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We use hardware to interact with programs.



The kernel **shares resources and controls access**.

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Linux

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- **1991:** Linus Torvalds announces his kernel project.
- Linux is a **Unix-like kernel**, distributed as free and open source software.
- A distribution combines the kernel with tools, libraries and applications.

Our working environment: a Linux system with the **Bash shell** and command-line tools.

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Unix, GNU and Linux

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Name	Meaning
Unix	The operating-system family that inspired Linux
GNU	The project behind Bash and many command-line tools
Linux	The kernel: manages processes, memory and hardware

GNU = GNU's Not Unix.
The name contains itself: a recursive acronym.

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Touch, click or type

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You already use interfaces to start programs.

Your action	Interface	Example
Touch a phone screen	Graphical interface	Open an app from its launcher
Click with a mouse	Graphical interface	Open a folder in a file manager
Type a command	Text interface	Ask Bash to list a directory

A smartphone can be a **terminal**: an endpoint for using a service. Its touchscreen, like a mouse or keyboard, provides **input**.

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Our terminal for this course?

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DEC VT100 · IF-3-SYS, G. Salagnac, INSA Lyon

A **physical terminal** sends keystrokes to a computer and displays its replies. The programs run on the connected computer.

Good news: yours fits in a window.

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Terminal, shell and command

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Name	Role
Terminal emulator	A window for keyboard input and text output
Bash, the shell	Interprets the line and runs commands
`ls`, a command	Lists directory entries

Type `ls` → Bash runs `ls` → the terminal displays its output.

TTY = *teletypewriter*: early terminals printed on paper.
Unix keeps the name, even for emulated terminals.

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A command and its arguments

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```
ls -l /home/alice
```

Part	Role
`ls`	**Command**: list directory entries
`-l`	**Option**: change how `ls` lists them
`/home/alice`	**Operand**: the directory to list

Both `-l` and `/home/alice` are **arguments** passed to `ls`.

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Short and long options

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Form	Example	Meaning for GNU `ls`
Short: **one dash**	`ls -a`	Include hidden entries
Long: **two dashes**	`ls --all`	Same as `-a`
Combined	`ls -la`	`-l` and `-a` together

Some options take a value: `head -n 5 notes.txt` prints five lines.

A **bare `--` ends options**.
`ls -- -notes.txt` treats `-notes.txt` as a filename.

Each command defines its options. Check `--help` or `man`.

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Getting help

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```
man ls
info coreutils
apropos directory
help cd
```

- **Tab** completes a name or displays possible completions.
- **Up** recalls a previous command. `history` lists history.
- `clear` clears the display. In `man`, **q** quits.

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2. Files and directories

Where is my file, and what can I do with it?

20/56

One directory tree

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A **directory** is a folder: it contains names of files and other directories.

```
/
├── home/
│   └── alice/
│       └── notes.txt
└── etc/
```

The common root is /. The path `/home/alice/notes.txt` follows the tree.
Alice's **home directory** is `/home/alice`; the university may use a different path.

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Personal files and system data

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Linux separates personal files, configuration and application data.

Directory	Typical contents
`/home/alice`	Alice’s personal files
`/etc`	System configuration
`/var`	Changing data: logs, caches, application state
`/var/lib`	Persistent application state
`/var/log`	Log files

`/var/lib` and `/var/log` are subdirectories of `/var`.

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Users, administrators and software

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Directory	Typical role
`/home`, `/root`	User homes; the root account's home
`/usr`	Most distribution-provided tools and libraries
`/usr/local`	Software installed locally by the administrator
`/opt`	Add-on software packages

`/` is the filesystem root.
`root` is also the administrator account's name.

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Temporary files and system information

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Directory	Purpose / lifetime
`/var`	Changing data, often persistent
`/tmp`	Temporary files; cleanup policy varies
`/run`	Runtime state, cleared at boot
`/dev`	Device interfaces
`/proc`, `/sys`	Views of processes, kernel and devices

A path can expose live system information.
It need not refer to ordinary data stored on disk.

24/56

Several filesystems, one tree

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```
/
├── home/       user files
├── mnt/        a temporary mount point
└── media/      removable media mount points
```

Mounting makes another filesystem visible at a directory.
The device node might be `/dev/sdb1`; its files appear at the mount point.

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The current directory

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Notation	Meaning
`.`	Current working directory
`..`	Parent directory
`/`	Root directory
`~`	Your home directory, expanded by the shell
`$HOME`	Variable containing your home path
`$PWD`	Variable containing the current directory path

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Absolute and relative paths

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Current directory: `/home/alice`

Path	Interpretation
`/home/alice/notes.txt`	Absolute: start at `/`
`notes.txt`	Relative: start in the current directory
`./notes.txt`	The same relative path
`../bob/notes.txt`	Go to the parent, then into `bob`

A path tells us **where to look**. Permissions determine whether access is allowed.

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Moving around

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```
pwd
cd /home/alice
cd ..
cd
```

- `pwd` prints the current working directory.
- `cd path` changes the current directory.
- `cd` or `cd ~` returns to your home.

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Listing directory entries

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```
ls
ls -l
ls -a
ls -la /home/alice
```

- `-l` adds details, including ownership and permissions.
- `-a` includes names beginning with `.`, such as `.bashrc`.
- `ls directory` lists that directory's contents.

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Creating and removing directories

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```
mkdir practice
rmdir practice
```

`rmdir` removes an **empty** directory.

`rm -r practice` also removes the directory's contents.
Shell deletion does **not** send files to a recycle bin.

30/56

Creating, copying and moving files

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```
touch notes.txt
cp notes.txt notes-copy.txt
mv notes-copy.txt archive.txt
rm archive.txt
```

`touch` creates a missing file, or updates timestamps of an existing file.

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Reading text files

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Command	Result
`cat notes.txt`	Print the whole file
`less notes.txt`	Browse interactively, **q** to quit
`more notes.txt`	Browse a page at a time
`head -n 5 notes.txt`	Print the first five lines
`tail -n 5 notes.txt`	Print the last five lines
`diff old.txt new.txt`	Show differences between two files

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Why is less called less?

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Mark Nudelman was investigating errors in large logs.
He wanted to read the lines **before** an error.

His version of `more` could not move backward.
The new pager's name joked about a **backward more**.

```
less app.log
```

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Concatenating files

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```
cat first.txt second.txt
cat
cat first.txt - second.txt
```

- With filenames, `cat` prints their contents **in argument order**.
- Without filenames, `cat` reads **standard input**.
- A standalone `-` means **read standard input here**.
- At an empty terminal input line, **Ctrl+D** indicates end of input.

The third command reads `first.txt`, then your keyboard input, then `second.txt`.

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A space can change the arguments

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We want to read **one file** named `meeting notes.txt`.

Without quotes

```
cat meeting notes.txt
```

`cat` receives **two filenames**:

- `meeting`
- `notes.txt`

With quotes

```
cat "meeting notes.txt"
```

`cat` receives **one filename**:

- `meeting notes.txt`

Quotes keep the filename together. The surrounding `"` characters are **not passed to `cat`**.

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Three ways to protect a space

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Each command passes the same filename: **`meeting notes.txt`**.

Form	What you type
Single quotes	`cat 'meeting notes.txt'`
Double quotes	`cat "meeting notes.txt"`
Backslash	`cat meeting\ notes.txt`

The surrounding quotes and the backslash protect the space. They are **not part of the filename**.

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Patterns select matching filenames

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The directory contains only **a.txt** and **meeting notes.txt**.

```
cat *.txt
```

The shell finds the matches, then passes **two filenames** to cat: a.txt and meeting notes.txt.

Pattern	Matches
*	Zero or more characters
?	Exactly one character
[abc]	One character from the set

The space **inside a matched filename** stays intact.

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Quotes keep a pattern literal

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Same directory: **a.txt** and **meeting notes.txt**.

Let the shell find matches

```
cat *.txt
```

cat receives **two filenames**:

- a.txt
- meeting notes.txt

It reads both files.

Protect the star with quotes

```
cat '*.txt'
```

cat receives **one filename**:

- *.txt

No file has that literal name here: cat reports an error.

cat "*.txt" and cat *.txt also pass a literal *.

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Single quotes or double quotes?

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`HOME` stores your home directory, here `/home/alice`. `$HOME` asks for its value; `echo` prints its arguments.

What you type	What you see
`echo '$HOME'`	`$HOME`
`echo "$HOME"`	`/home/alice`

Single quotes keep `$HOME` literal. **Double quotes** allow it to expand. Both protect spaces and filename pattern characters such as `*`.

With double quotes, we can use the variable and protect the space:

```
cat "$HOME/meeting notes.txt"
```

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Checkpoint: files and paths

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1. What does `cd ..` do?
2. Which `ls` option includes hidden entries?
3. What does `cat a.txt b.txt` print first?
4. Which filename does `cat '*.txt'` try to open?

1. Move to the parent directory.
2. `-a`
3. The contents of `a.txt`.
4. The literal filename `*.txt`. The shell does not expand the star.

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3. Who may access this file?

Why can I read this file, but cannot change or delete it?

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Users, groups and ownership

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```
whoami
id
ls -ld .
ls -l
```

- `id` shows your user identity and groups.
- `ls -l` shows the file's owner, group and permission bits.
- `ls -ld` inspects a directory itself.

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Permission classes

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```
-rwxr-xr--
  u  g  o
```

Class	Applies to
u	The file's owner
g	Members of the file's group
o	Other users
a	All three classes when using `chmod`

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What r, w and x allow

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Bit	Regular file	Directory
r	Read contents	List entry names
w	Modify contents	Add or remove entries*
x	Execute	Search / traverse

*Changing directory entries also requires search permission. Other restrictions, such as the sticky bit, may apply.

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Changing permission bits

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```
chmod u+x script.sh
chmod go-w notes.txt
chmod u=rw,g=r,o= notes.txt
```

- `+` adds the selected permission bits.
- `-` removes them.
- `=` sets the selected class to the listed permissions.

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Numeric permissions

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Each class adds its bit values: **r = 4**, **w = 2**, **x = 1**.

Owner	Group	Others
`rwx` = **7**	`rw-` = **6**	`r--` = **4**

```
chmod 764 notes.txt
```

The three digits describe **owner, group, others**, in that order.

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Checkpoint: permissions

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1. Decode `640` for a regular file.
2. Add execute permission for the owner of `run.sh`.
3. Which directory bit allows traversal?

1. Owner: `rw-`
   Group: `r--`
   Others: `---`
2. `chmod u+x run.sh`
3. `x`

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“I can see the name, but cannot read it”

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```
# Illustrative server metadata
 drwxr----- alice students  shared/
 -rw-r----- alice students  shared/notes.txt
```

Bob belongs to `students`. He can list the names, but cannot open `notes.txt`. Why?

The directory’s group bits lack **x**: traversal.
If team access is intended, the owner can add `g+x`.

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CM1 summary: files and permissions

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1. Locate configuration, software, data and logs.
2. List and read files; quote names containing spaces.
3. Check owner, group and other permissions.

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Optional reference

Examples to revisit during TD or personal practice

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Operating system, shell and GUI

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Layer	What it gives us
Operating system	Processes, storage and access control
Graphical interface	Windows, menus and direct visual interaction
Shell	Commands we can combine and save in scripts
Application	A task-specific tool, such as Excel or a database

Your laptop can run Windows while the company's application runs on Linux. You need to work with both.

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Path components and word counts

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```
dirname /home/alice/notes.txt
basename /home/alice/notes.txt
wc -w notes.txt
```

- `dirname` prints `/home/alice`.
- `basename` prints `notes.txt`.
- `wc -w` counts **words**. `wc -l` counts newline characters.

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Hard links and symbolic links

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Hard link

```
ln notes.txt other.txt
```

- Another name for the **same file**.
- Removing one name leaves the other usable.
- Same filesystem, normally no directory links.

Symbolic link

```
ln -s notes.txt shortcut
```

- Stores a **path** to a target.
- May cross filesystems or target a directory.
- Can become dangling if the target path disappears.

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Builtins and external commands

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Builtin

Implemented inside the shell.

```
cd ..
```

Can change the current shell's state.

External program

Loaded from an executable file.

```
ls -l
```

The shell searches directories in PATH.

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Recognising a command

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```
type cd
type ls
type -a echo
help cd
```

`type` reports how Bash resolves a name.
`help` documents its builtins.

Seeing each argument with printf

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Only two files: **`a.txt`** and **`meeting notes.txt`**.

`printf '<%s>\n'` displays each following argument on its own line.

```
printf '<%s>\n' *.txt
```

```
<a.txt>
<meeting notes.txt>
```

Two filename arguments.

```
printf '<%s>\n' '*.txt'
```

```
<*.txt>
```

One literal argument.